



```
In [7]: year = 2100

if year % 400 == 0:
    print(True)

elif year % 100 == 0:
    print(False)

elif year % 4 == 0:
    print(True)

else:
    print(False)
```

False

```
In [1]: # Simple Calculator

while(True):
    print('-----')
    a = int(input('Enter First Number : '))
    b = int(input('Enter Second Number : '))

    print('\n--MENU OPTIONS--')
    print('+ -> Addition')
    print('- -> Subtraction')
    print('* -> Multiplication')
    print('/ -> Division')

    o = input('\nEnter your choice (To stop enter BREAK) : ')
    res = None

    if(o == 'BREAK'):
        break

    elif(o == '+'):
        res = a+b

    elif(o == '-'):
        res = a-b

    elif(o == '*'):
        res = a*b

    elif(o == '/'):
        res = a/b

    else:
        res=None

    if(res == None):
        print('\n--ERROR--')

    else:
```

```
print('Result : ',res)

print('\n')
```

```
-----
--MENU OPTIONS--
+ -> Addition
- -> Subtraction
* -> Multiplication
/ -> Division
```

In [2]: *# Stack Implementation*

```
stack = list()

def sprint():
    global stack
    print('The Elements of Stack are : ', stack)

def push():
    global stack
    ele = int(input('Enter an element to push into stack : '))
    stack.append(ele)

def pop():
    global stack
    ele = stack[-1]
    stack = stack[:-1]
    print('The popped element is : ', ele)
```

In [3]:

```
while(True):
    print('-----')

    print('--MENU--')
    print('1. Print the Stack')
    print('2. Push New Element')
    print('3. Pop An Element')
    print('0. Stop')

    o = int(input('\nEnter your option : '))

    if(o == 1):
        sprint()

    elif(o == 2):
        push()

    elif(o == 3):
        pop()

    elif(o == 0):
        break
```

```
else:
    print('-- INVALID --')
```

```
-----
--MENU--
1. Print the Stack
2. Push New Element
3. Pop An Element
0. Stop
The Elements of Stack are :  []
-----
--MENU--
1. Print the Stack
2. Push New Element
3. Pop An Element
0. Stop
-----
--MENU--
1. Print the Stack
2. Push New Element
3. Pop An Element
0. Stop
The Elements of Stack are :  [8]
-----
--MENU--
1. Print the Stack
2. Push New Element
3. Pop An Element
0. Stop
-----
--MENU--
1. Print the Stack
2. Push New Element
3. Pop An Element
0. Stop
The Elements of Stack are :  [8, 9]
-----
--MENU--
1. Print the Stack
2. Push New Element
3. Pop An Element
0. Stop
-----
--MENU--
1. Print the Stack
2. Push New Element
3. Pop An Element
0. Stop
```

The Elements of Stack are : [8, 9, 10]

--MENU--

1. Print the Stack
2. Push New Element
3. Pop An Element
0. Stop

The popped element is : 10

--MENU--

1. Print the Stack
2. Push New Element
3. Pop An Element
0. Stop

The Elements of Stack are : [8, 9]

--MENU--

1. Print the Stack
2. Push New Element
3. Pop An Element
0. Stop

In []: